

DISCLAIMER: AI-ASSISTED LEARNING PROJECT

This presentation and the accompanying repository are educational projects created for learning and experience.

Large Language Models (AI) were used extensively to aid in completing, writing, and architecting this analysis pipeline and its documentation.

SURFACE PROXY · SPATIAL ANALYSIS · OPEN WORKFLOW

Surface titanium and lunar crustal magnetism

- Global-map analysis over the declared joint-valid footprint (LROC $\pm 70^\circ$ and Imbrian-age restriction). **It does not directly test** the dated temporal mechanism of Nichols et al. (2026).



THE PUZZLE

A magnetized crust with no magnet behind it

- **No global field today**

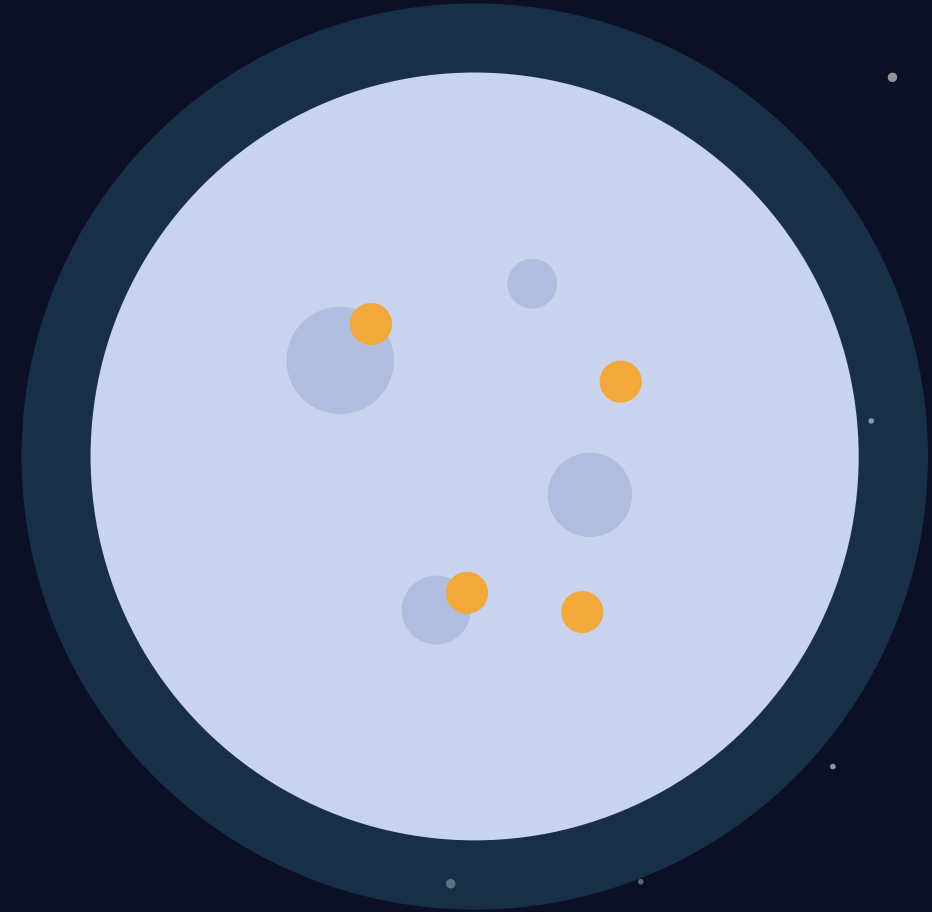
Unlike Earth, the Moon has no active core dynamo — so it should be magnetically dead.

- **Yet the crust is magnetized**

Orbiting spacecraft measure strong, patchy magnetic anomalies locked into ancient rock.

- **Timing is the missing variable**

Dated samples constrain ancient field evolution. Present-day surface maps test spatial co-location, not the temporal dynamo mechanism.



THE SCOPE

One temporal mechanism motivates one spatial proxy test

● Motivation · dated samples

Nichols, Wade & Stephenson (2026), Nature Geoscience

Dated palaeointensity and Ti-bearing samples motivate a time-dependent core-mantle boundary melting and dynamo hypothesis.

Temporal mechanism

● Benchmark · impact antipodes

Hood & Artemieva (2008), Icarus

Fold-matched H2 PR-AUC .110579 vs null mean .031970, SD .026728 and 95th .106832 ($p=.039604$).

100 accepted shifts · 5/5 folds evaluable

Does surface TiO_2 add spatial information for crustal magnetism after mapped geologic controls?

1

Repository plan

Success and failure criteria are fixed in the plan and code the pipeline runs.

2

Power-gated

Positive and negative outcomes are explicit; negative claims require adequate detection power.

3

Scope-limited

The result concerns spatial co-location—not Nichols et al.'s temporal dynamo mechanism.

THE EVIDENCE

Five global datasets, plus a mapped mare-proxy mask

- **Magnetic field (what we predict)**

Kaguya + Lunar Prospector surface vector map

- **Surface titanium — TiO_2 (proxy)**

LROC WAC UV/Vis abundance map; mare-derived calibration

- **Bouguer gravity**

GRAIL GRGM1200A — subsurface density structure

- **Crustal thickness**

GRAIL crustal-thickness archive

- **Geology + mare proxy**

USGS units define age and mapped mare-proxy scope

~2 GB

of institutional data

23

pinned data products

SHA-256

every byte verified

1° grid

≈ 30 km at the equator

THE METHOD

Four stages separate description from inference

1

Align

Align five maps and the derived mapped mare-proxy mask on one equal-area lunar grid.

2

Describe

Fit continuous amplitude with controls, then add Ti features: raw + 25/50/100 km.

3

Restrict

Repeat controls vs. controls+raw-TiO₂ only inside the mapped USGS mare-proxy mask.

4

Calibrate

Match each statistic to its own spatial null, then inject H₂ signals to measure power.

Why blocking still falls short: 30° (~910 km) and 60° (~1819 km) holdouts remain below the ~3752 km dependence range. Both nulls used the first 100 eligible 5/5-fold unique nonidentity shifts from 125 candidates; 12 four-fold candidates were rejected. Accepted prevalence: .007686–.022767.

Reproducible evidence still has design limits

● Analysis plan and code

The plan and parameter file are version-controlled separately from the generated results.

● Logical-fallacy audit

Inference traps are documented and corrected in the open—including the matched H2 null.

● Fail-closed provenance

Hashes, real/synthetic guards and mare-mask alignment are enforced in code.

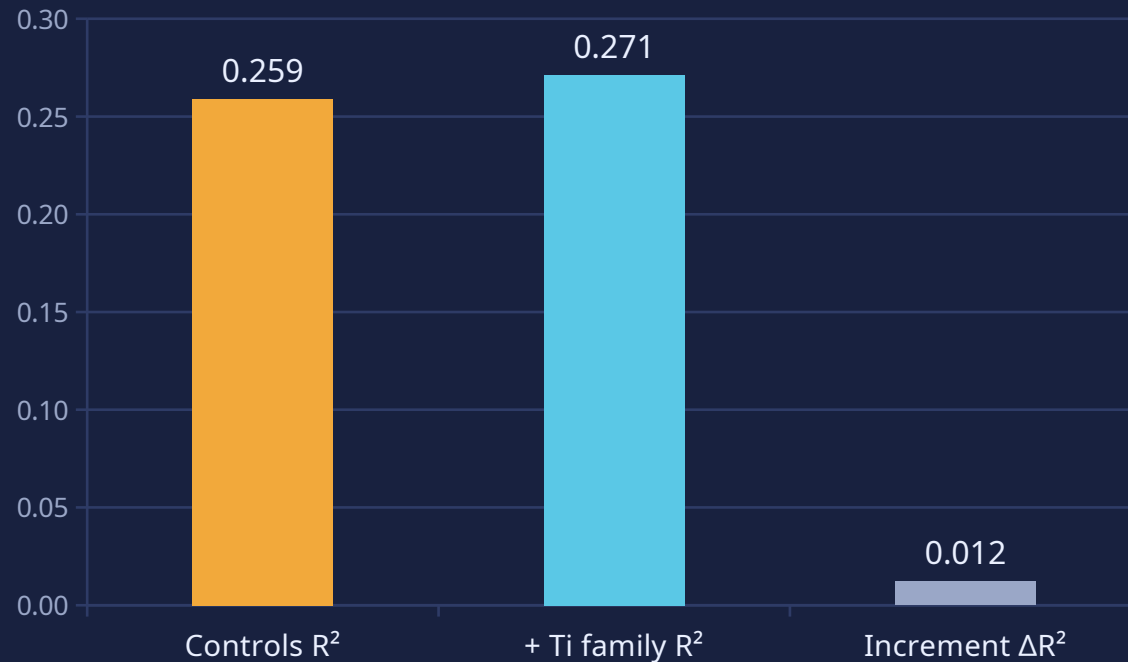
● Automated tests

Guards cover provenance, terrain validity, power gating and report claims.

THE RESULT

The full Ti family adds only a small descriptive increment

Held-out continuous R^2 (descriptive)



$\Delta R^2 +.0121$

controls .2590 $\rightarrow$.2712 with full Ti family; descriptive only

$\Delta \text{PR-AUC}$
 $-.002992$

remove raw/buffered Ti + Ti $\times$ gravity interactions; $p=.78125$

$n_{\text{eff}} \approx 1$

30° 910 km holdout < 3752 km dependence range

STATUS: INCONCLUSIVE_LOW_POWER

This spatial proxy result neither confirms nor refutes Nichols et al.'s temporal mechanism.

Mapped mare-proxy TiO_2 is positive but not significant

Mapped mare-proxy binary

$\Delta\text{PR-AUC} +.067614$; $+\text{TiO}_2 .125209$ versus controls $.057595$ ($p=.21875$).

$n=6,232 \cdot 58$ positives



Mapped mare-proxy continuous

Controls $R^2 .425691 \rightarrow .438387$; raw-Ti increment $+.012696 \pm .012575$. Common-fold gain vs full raw-Ti: $+.001445$.

proxy range 3289 km $\cdot n_{\text{eff}}=1$

Interpretation: Both proxy checks remain descriptive: the mapped proxy footprint still supplies only one dependence-scale unit, not **an independent global confirmation**.

Three diagnostics set the inference ceiling

- **Dependence exceeds blocks**

$n_{\text{eff}} \approx 1$. Even 60° blocks (~1819 km) remain below the ~3752 km dependence range.

- **Matched H2 benchmark recovered**

Full PR-AUC .088972 vs fold-matched null mean .112697, SD .056903 and 95th .225555 ($p=.564356$); H2 match $p=.039604$.

- **Power floor is not adequate power**

H2 recovery at $\beta=0/.5/1/1.5/2/3-4$: 3/30, 18/30, 27/30, 28/30, 29/30, 30/30. Point MDE $\beta=1$; Wilson-lower MDE $\beta=2$.

“

**INCONCLUSIVE
LOW POWER**

$\beta=1$ implies risk +.0557 and $OR \approx 607$; no justified target effect.

The next test must be time-resolved.

- A null here is failure-to-detect, not disproof: a surface-map co-location study, not a temporal dynamo test.
- Use dated samples or time-resolved proxies to test the Nichols mechanism directly.
- Expand calibrated TiO_2 coverage and use dependence-scale blocks before any negative claim.



Thank you · github.com/lemoon01110/Lunar-Titanium-Magnetism